



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 13/09/2018*

PROBLEM

We have a random variable X having uniform distribution between $-\theta$ and θ , where $\theta > 0$.

Assuming that we observe N independent realizations of X :

- (1) Plot the 1st order pdf of X and then derive and plot the likelihood function $L(\theta)$.
- (2) Derive a consistent estimator of parameter θ by applying the method of moments (MM). If possible, derive its bias and mean square error (MSE) and establish if the estimator is consistent.
- (3) Derive the maximum likelihood (ML) estimator of parameter θ , its bias and mean square error (MSE), and establish if the estimator is consistent.
- (4) Comment on the pros and cons of the estimators previously derived.



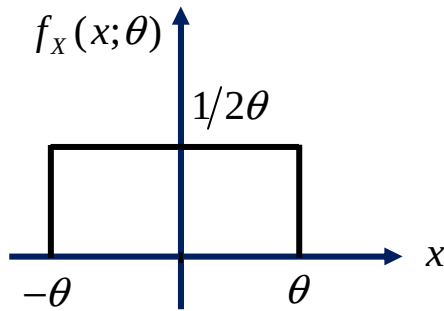
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Solution:

(1) The 1st order pdf is given by

$$f_X(x; \theta) = \frac{1}{2\theta} \text{rect}\left(\frac{x}{2\theta}\right) = \begin{cases} 1/2\theta, & -\theta \leq x \leq \theta \\ 0, & \text{otherwise} \end{cases}$$



Since the observed data are independent and identically distributed (IID), the likelihood function is given by the product of the marginal pdf's:

$$\begin{aligned} L(\theta) &= \prod_{i=1}^N f_X(x_i; \theta) = \prod_{i=1}^N \frac{1}{2\theta} \text{rect}\left(\frac{x_i}{2\theta}\right) = \frac{1}{(2\theta)^N} \prod_{i=1}^N \text{rect}\left(\frac{x_i}{2\theta}\right) \\ &= \begin{cases} \frac{1}{(2\theta)^N}, & -\theta \leq x_1 \leq \theta, -\theta \leq x_2 \leq \theta, \dots, -\theta \leq x_N \leq \theta \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} \frac{1}{(2\theta)^N}, & |x_1| \leq \theta, |x_2| \leq \theta, \dots, |x_N| \leq \theta \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

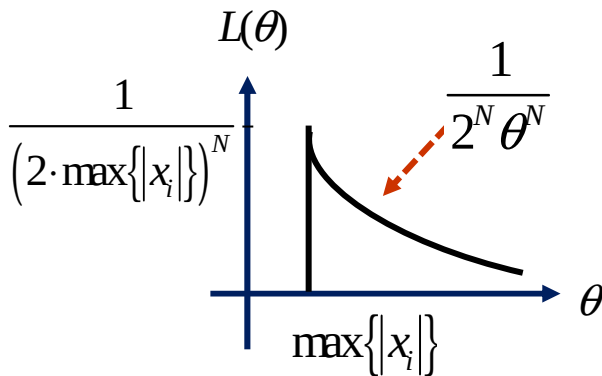
which can be expressed as:

$$L(\theta) = \begin{cases} \frac{1}{(2\theta)^N}, & \max\{|x_i|\} \leq \theta \\ 0, & \text{otherwise} \end{cases}$$



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(2) The mean of X is 0 so we cannot use it to estimate θ , we should resort to the variance to find a Method of Moments (MM) estimator:

$m_{1X} = E\{X\} = 0$ (the mean value is the symmetry point of the pdf)

$$\mu_2 = E\{(X - m_1)^2\} = E\{X^2\} = \frac{(2\theta)^2}{12} = \frac{\theta^2}{3} \Rightarrow \theta = \sqrt{3\mu_2}$$

$$\hat{\theta}_{MM1}(\mathbf{x}) = \sqrt{3\hat{\mu}_2} = \sqrt{\frac{3}{N} \sum_{i=1}^N X_i^2}$$

We can immediately say that the estimator is non-linear and biased, but consistent. However, it is difficult to find the expressions of bias and MSE in closed form for finite sample size, i.e. as a function of N .

Another possibility to find an estimator by using the MM is based on the observation that $Y \triangleq |X|$ is a random variable uniformly distributed between 0 and θ , so we have: $E\{Y\} = E\{|X|\} = \theta/2$. As a consequence, a possible estimator of θ based on the MM can be derived as follows:



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$$\theta = 2m_{1Y} \Rightarrow \hat{\theta}_{MM2}(\mathbf{x}) = 2\hat{m}_{1Y} = \frac{2}{N} \sum_{i=1}^N Y_i = \frac{2}{N} \sum_{i=1}^N |X_i|$$

$$E\{\hat{\theta}_{MM2}(\mathbf{x})\} = E\left\{\frac{2}{N} \sum_{i=1}^N |X_i|\right\} = \frac{2}{N} \sum_{i=1}^N E\{|X_i|\} = \frac{2}{N} \sum_{i=1}^N \frac{\theta}{2} = \theta$$

$$MSE\{\hat{\theta}_{MM2}(\mathbf{x})\} = \text{var}\{\hat{\theta}_{MM2}(\mathbf{x})\} = \text{var}\left\{\frac{2}{N} \sum_{i=1}^N |X_i|\right\} = \frac{4}{N^2} \sum_{i=1}^N \text{var}\{|X_i|\} = \frac{4}{N^2} \cdot N \cdot \frac{\theta^2}{12} = \frac{\theta^2}{3N}$$

This MM estimator, called MM2, is unbiased and consistent.

(3) In this case, the likelihood function is not continuous and differentiable in the definition interval of the parameter θ , that is $\theta \in (0, +\infty)$. Hence, the ML estimate cannot be obtained by solving the likelihood equation, but it must be obtained by looking directly for the value of θ for which $L(\theta)$ is maximum. In our case, by inspection of the previous figure we immediately obtain:

$$\hat{\theta}_{ML} = \arg \max_{\theta} L(\theta) = \max\{|x_1|, |x_2|, \dots, |x_N|\}$$

Following the usual derivation, (see course notes Ch. 4 pages 66-69) we find:

$$f_{\hat{\theta}_{ML}}(y) = \frac{N}{\theta} \left(\frac{y}{\theta}\right)^{N-1} \text{rect}\left(\frac{y - \theta/2}{\theta}\right)$$

$$E\{\hat{\theta}_{ML}\} = \frac{N\theta}{N+1}, \quad E\{\hat{\theta}_{ML}^2\} = \frac{N\theta^2}{N+2} \Rightarrow \text{var}\{\hat{\theta}_{ML}\} = \frac{N\theta^2}{(N+1)^2(N+2)}$$

$$b\{\hat{\theta}_{ML}\} = E\{\theta - \hat{\theta}_{ML}\} = \frac{\theta}{N+1} \xrightarrow{N \rightarrow \infty} 0$$

$$MSE\{\hat{\theta}_{ML}(\mathbf{x})\} = \left(b\{\hat{\theta}_{ML}\}\right)^2 + \text{var}\{\hat{\theta}_{ML}\} = \frac{2\theta^2}{(N+1)(N+2)} \xrightarrow{N \rightarrow \infty} 0$$

Hence, the ML estimator is biased but consistent.



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$$(4) \quad \hat{\theta}_{MM1}(\mathbf{x}) = \sqrt{\frac{3}{N} \sum_{i=1}^N X_i^2}, \quad \hat{\theta}_{MM2}(\mathbf{x}) = \frac{2}{N} \sum_{i=1}^N |X_i|, \quad \hat{\theta}_{ML} = \max \{|x_1|, |x_2|, \dots, |x_N|\}$$

Performance comparison:

The MSE of the ML estimator goes to zero with the law $1/N^2$. Instead, the MSE of the MM2 estimator goes to zero with the law $1/N$. Hence, the ML estimator is much more efficient. The performance of the MM1 estimator should be evaluated by Monte Carlo simulation. The MM1 estimator should have the same asymptotical behavior as the MM2 estimator.

Comparison in terms of computational complexity:

The ML estimator is **non-linear**, it requires calculation of the maximum of the absolute values of a set of N data (data ordering).

The MM1 and MM2 estimators are **non-linear**, but a linear function of the squares or of the absolute values of the data samples, hence both of them are much more efficient from the point of view of **computational complexity**.

The choice between them depends on the constraints imposed by the specific application: if estimation accuracy is the most important issue, then we should use the ML estimate; if instead it is more important to keep low the computational complexity, we should use the MM2 estimator.